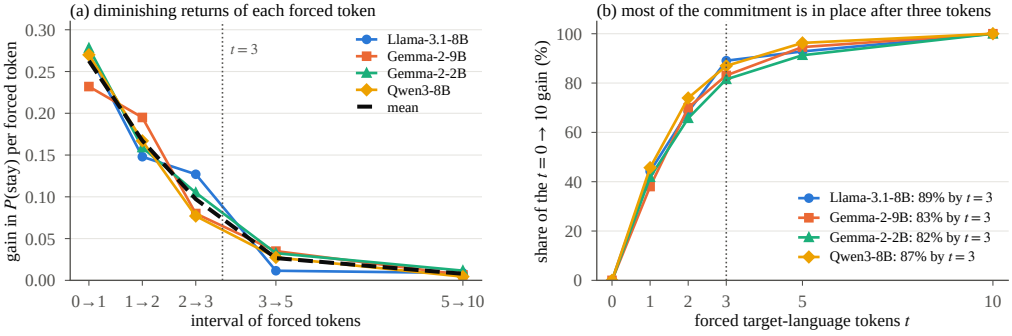
(a) diminishing returns of each forced token
gain in P(stay) per forced token
0.30
0.25
0.20
0.15
0.10
0.05
0.00
t = 3
Llama-3.1-8B
Gemma-2-9B
Gemma-2-2B
Qwen3-8B
mean
0→1
1→2
2→3
3→5
5→10
interval of forced tokens
(b) most of the commitment is in place after three tokens
share of the t = 0 → 10 gain (%)
100
80
60
40
20
0
Llama-3.1-8B: 89% by t = 3
Gemma-2-9B: 83% by t = 3
Gemma-2-2B: 82% by t = 3
Qwen3-8B: 87% by t = 3
0
1
2
3
5
10
forced target-language tokens t